

~~Wieder~~
für ~~Wieder~~.

$$1) \int f(x) dx = F(x) + C \Rightarrow \int f(ax+b) dx = \frac{1}{a} F(ax+b) + C \quad (a \neq 0)$$

$$2) \quad \begin{array}{l} t = g(x) \\ \int f(g(x)) \cdot g'(x) dx = \int f(t) dt. \end{array}$$

$$3) \int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$4) \int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| + C$$

$$5) \int (ax+b)^r dx = \frac{1}{a} \cdot \frac{1}{r+1} (ax+b)^{r+1} + C$$

$$6) \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$7) \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$8) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$